Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the range of the data set: 20, 39, 10, 12, 26.

Final Answer: 29. Range $= \max - \min = 39 - 10 = 29$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the median of the data set: 32, 10, 29, 13, 37, 36, 34, 38.

Final Answer: 33. Sorted: 10, 13, 29, 32, 34, 36, 37, 38. The mean of the two middle values $\frac{32+34}{2} = 33$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two independent events A and B have $P(A) = \frac{3}{4}$ and $P(B) = \frac{9}{10}$. Find $P(A \text{ and } B)$.

Final Answer: $\frac{27}{40}$. Independent events multiply: $P(A) \times P(B) = \frac{3}{4} \times \frac{9}{10} = \frac{27}{40}$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 5 red marbles, 4 blue marbles, and 3 green marbles. Two marbles are drawn at random one after another without replacement. What is the probability that both marbles are red?

Total number of marbles initially: $5 + 4 + 3 = 12$. Probability that the first marble is red: $P(R_1) = \frac{5}{12}$. Since it is drawn without replacement, 11 marbles remain, 4 of which are red. Probability that the second marble is red: $P(R_2|R_1) = \frac{4}{11}$. Using the multiplication rule: $P(R_1 \cap R_2) = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \mathbf{\frac{5}{33}}$

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The scores on a statewide standardized test are normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. Approximately what percentage of test takers score between 400 and 700?

Express the boundaries in terms of standard deviations from the mean: $400 = \mu - 1\sigma$ $700 = \mu + 2\sigma$ Using the Empirical Rule: Between $\mu - 1\sigma$ and μ: 34.1%. Between μ and $\mu + 1\sigma$: 34.1%. Between $\mu + 1\sigma$ and $\mu + 2\sigma$: 13.6%. Total percentage: $34.1\% + 34.1\% + 13.6\% = \mathbf{81.8\%}$ (or approximately **81.5% – 82%**).

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set contains the values: 4, 7, 7, 9, 12, 15. What are the mean, median, and range of this data set?

Sum: $4 + 7 + 7 + 9 + 12 + 15 = 54$.Mean: $\bar{x} = \frac{54}{6} = \mathbf{9}$.Median: The two middle values are 7 and 9, so Median $= \frac{7+9}{2} = \mathbf{8}$.Range: Max $-$ Min $= 15 - 4 = \mathbf{11}$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A committee of 3 students is to be chosen from a group of 8 candidates. How many different committees can be formed?

Because the order of committee members does not matter, this is a combination:

${}_8C_3 = \binom{8}{3} = \frac{8!}{3!\,(8-3)!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = \mathbf{56}$ different committees

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bivariate regression model for student preparation hours x and exam grade y yields the line of best fit $\hat{y} = 4.8x + 52$. A student who studied for 8 hours scored 94 on the exam. (a) What is the model's predicted score for this student? (b) Compute and interpret the student's regression residual.

(a) Predicted Score $\hat{y}$: Substitute $x = 8$ into the regression equation:
$\hat{y} = 4.8(8) + 52 = 38.4 + 52 = \mathbf{90.4}$. (b) Residual Calculation & Interpretation:
$\text{Residual} = y - \hat{y} = 94 - 90.4 = \mathbf{+3.6}$. Because the residual is positive (+3.6), the student's actual score was 3.6 points higher than the score predicted by the regression line (the model underestimated the student's performance).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A student has scores of 78, 85, 92, and 81 on four exams. What score must the student earn on the fifth exam to achieve an overall arithmetic mean of 85 across all five exams?

Step 1: Determine the required sum of all 5 exam scores: $\sum x = 5 \times 85 = 425$. Step 2: Sum the first four exam scores: $78 + 85 + 92 + 81 = 336$. Step 3: Subtract the accumulated total from the required total to find the 5th score x_5: $x_5 = 425 - 336 = \mathbf{89}$. The student must score 89 on the fifth exam.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The equation of a circle is given by $x^2 + y^2 + 10x - 4y + 20 = 0$. What is the radius of this circle?

Group terms and complete the square for x and y: $(x^2 + 10x) + (y^2 - 4y) = -20$ Half the coefficients squared: $(10/2)^2 = 25$, $(-4/2)^2 = 4$. Add $25 + 4 = 29$ to both sides:
$(x^2 + 10x + 25) + (y^2 - 4y + 4) = -20 + 25 + 4 = 9$ $(x + 5)^2 + (y - 2)^2 = 9$ Since $r^2 = 9$, the radius is $r = \sqrt{9} = \mathbf{3}$.

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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Each interior angle of a regular polygon measures 150°. How many sides does this polygon have?

Using the exterior angle method (much faster on the ACT than interior angles): Each exterior angle is supplementary to the interior angle: $180^\circ - 150^\circ = 30^\circ$. Since the exterior angles of any regular polygon sum to 360°: $n = \frac{360^\circ}{30^\circ} = \mathbf{12}$ sides (dodecagon)

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An infinite geometric sequence has first term $a_1 = 18$ and common ratio $r = -\frac{1}{3}$. What is the sum of this infinite series?

Since $|r| = |-\frac{1}{3}| < 1$, the infinite series converges. Using the formula $S_\infty = \frac{a_1}{1-r}$:
$S_\infty = \frac{18}{1-(-1/3)} = \frac{18}{1+1/3} = \frac{18}{4/3} = 18 \times \frac{3}{4} = \frac{54}{4} = \mathbf{\frac{27}{2}}$ (or 13.5)

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following expressions is equivalent to $\frac{1-\sin^2\theta}{\cos\theta}$ for all values of θ where the expression is defined?

Using the fundamental Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$, we have $1 - \sin^2\theta = \cos^2\theta$. Substituting this into the numerator: $\frac{1-\sin^2\theta}{\cos\theta} = \frac{\cos^2\theta}{\cos\theta} = \mathbf{\cos\theta}$

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $\cos\theta = -\frac{4}{5}$ and θ lies in Quadrant II, what is the value of $\tan\theta$?

In Quadrant II, $x = -4$ and hypotenuse $r = 5$. Using $x^2 + y^2 = r^2$:
$(-4)^2 + y^2 = 5^2 \Rightarrow 16 + y^2 = 25 \Rightarrow y = 3$ (positive in QII) Therefore: $\tan\theta = \frac{y}{x} = \frac{3}{-4} = \mathbf{-\frac{3}{4}}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

If $A = (2, -1; 3, 4)$ and $B = (5; -2)$, what is the matrix product AB?

The product of a 2×2 matrix and a 2×1 matrix is a 2×1 column vector. Compute row by column: First row: $2(5) + (-1)(-2) = 10 + 2 = 12$. Second row: $3(5) + 4(-2) = 15 - 8 = 7$. Thus $AB = (\mathbf{12; 7})$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Practice 1.1: A jacket costs 120 after a 25% discount. What was the original price before the discount?

Let original price = x. A 25% discount means the customer pays $100\% - 25\% = 75\% = 0.75$ of the original price: $0.75x = 120 \Rightarrow x = \frac{120}{0.75} = \frac{120}{3/4} = 120 \times \frac{4}{3} = 160$. The original price was 160.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Line K passes through the points $(-3, 5)$ and $(1, -3)$. What is the slope of a line perpendicular to line K?

First calculate the slope of line K: $m_K = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3-5}{1-(-3)} = \frac{-8}{4} = -2$ A line perpendicular to K has slope $m_\perp = -\frac{1}{m_K}$: $m_\perp = -\frac{1}{-2} = \mathbf{\frac{1}{2}}$

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

A right circular cylinder has radius 3 cm and height 8 cm. What is its total surface area in terms of π?

The total surface area includes the two circular bases and the lateral curved surface:

$\text{TSA} = 2\pi r^2 + 2\pi rh = 2\pi(3^2) + 2\pi(3)(8) = 18\pi + 48\pi = \mathbf{66\pi}$ cm^2

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

What is the value of the complex expression $i^{47} - i^{18}$?

Divide exponents by 4: $47 = 4(11) + 3 \Rightarrow i^{47} = i^3 = -i$. $18 = 4(4) + 2 \Rightarrow i^{18} = i^2 = -1$. Therefore: $i^{47} - i^{18} = -i - (-1) = \mathbf{1 - i}$

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Plane Geometry

What is the equation of the circle with centre $(-2, 4)$ and radius 5?

Using the standard equation $(x-h)^2 + (y-k)^2 = r^2$ with center $(h, k) = (-2, 4)$ and $r = 5$:

$(x-(-2))^2 + (y-4)^2 = 5^2 \Rightarrow (x+2)^2 + (y-4)^2 = 25$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scored $82, 88, 91,$ and 85 on four exams. What score must the student earn on the fifth exam to achieve an overall mean score of 88?

a. 90

b. 92

c. 94 ✓

d. 96

Required total for 5 exams: $5 \times 88 = 440$. Current sum: $82 + 88 + 91 + 85 = 346$. Fifth score required: $440 - 346 = 94$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a set of 7 integers: $\{4, 7, 9, x, 14, 18, 20\}$ arranged in ascending order, the median is equal to the mean. What is the value of x?

a. 12 ✓

b. 13

c. 14

d. 15

Median of 7 sorted values is the 4th value: x. Mean $= \frac{4+7+9+x+14+18+20}{7} = \frac{72+x}{7}$. Equating: $7x = 72 + x \Rightarrow 6x = 72 \Rightarrow x = 12$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a chemistry course, homework counts for 20%, labs count for 30%, and tests count for 50%. If a student earns 90 on homework, 80 on labs, and 86 on tests, what is the weighted final grade?

a. 84

b. 85 ✓

c. 86

d. 87

Weighted grade $= 0.20(90) + 0.30(80) + 0.50(86) = 18 + 24 + 43 = 85$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set consists of the numbers $\{3, 7, 8, 12, 15, 18, 21\}$. What is the interquartile range (IQR) of this data set?

a. 9

b. 11 ✓

c. 12

d. 14

The median is 12. The lower half is $\{3, 7, 8\}$ with $Q_1 = 7$. The upper half is $\{15, 18, 21\}$ with $Q_3 = 18$. Thus IQR $= Q_3 - Q_1 = 18 - 7 = 11$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If every value in a data set is multiplied by 3 and then increased by 5, how is the standard deviation σ affected?

a. It increases by 5

b. **It is multiplied by 3** ✓

c. It is multiplied by 3 and increased by 5

d. It remains unchanged

Multiplying by a constant c scales standard deviation by $|c| = 3$. Adding a constant shifts the center but does not change the spread. Hence, standard deviation is multiplied by 3.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following data sets has a standard deviation equal to 0?

a. $\{5, 5, 5, 5, 5\}$ ✓

b. $\{1, 2, 3, 4, 5\}$

c. $\{-2, 0, 2\}$

d. $\{10, 20, 30\}$

A standard deviation of 0 occurs if and only if all values in the data set are identical.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 6 red marbles, 8 blue marbles, and 10 green marbles. If one marble is selected at random, what is the probability that it is NOT blue?

a. $\frac{1}{3}$

b. $\frac{2}{3}$ ✓

c. $\frac{1}{4}$

d. $\frac{3}{4}$

Total marbles $= 6 + 8 + 10 = 24$. Non-blue marbles $= 6 + 10 = 16$. Probability $= \frac{16}{24} = \frac{2}{3}$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two fair standard six-sided dice are rolled simultaneously. What is the probability that the sum of the two numbers rolled is greater than 9?

a. $\frac{1}{6}$ ✓

b. $\frac{5}{36}$

c. $\frac{1}{9}$

d. $\frac{7}{36}$

Outcomes with sum > 9 are sums of $10, 11, 12$: $(4,6), (5,5), (6,4), (5,6), (6,5), (6,6)$ = 6 outcomes. Probability $= \frac{6}{36} = \frac{1}{6}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.5$, what is the probability that at least one of the events occurs, $P(A \cup B)$?

a. 0.70 ✓

b. 0.80

c. 0.90

d. 0.20

For independent events, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.70$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many distinct ways can a committee of 4 members be chosen from a group of 9 people?

a. 36

b. 84

c. 126 ✓

d. 3024

Selection order does not matter: $\binom{9}{4} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many different 4-digit identification codes can be formed using digits 0 through 9 if repetition of digits is not allowed and the first digit cannot be 0?

a. $4,536$ ✓

b. $5,040$

c. $9,000$

d. $3,024$

Choices for 1st digit (1-9) $= 9$. Choices for 2nd digit (0-9 excluding 1st) $= 9$. Choices for 3rd digit $= 8$. Choices for 4th digit $= 7$. Total $= 9 \times 9 \times 8 \times 7 = 4,536$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A restaurant offers 4 appetizers, 6 main courses, and 3 desserts. How many distinct 3-course dinner combinations are possible?

a. 13

b. 36

c. 72 ✓

d. 144

By the fundamental counting principle: $4 \times 6 \times 3 = 72$ combinations.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set of 9 values has a mean of 54. A new value of 224 is added. What is the new mean?

a. 43

b. **71** ✓

c. 47

d. 138

Old total $= 9 \times 54 = 486$. New total $= 486 + 224 = 710$. New mean $= 710/10 = 71$.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A container holds only red, blue, and green marbles. Initially, the ratio of red marbles to blue marbles is $3:4$. First, 6 green marbles are added to the container, after which the probability of randomly drawing a green marble is $\frac{1}{3}$. Next, 4 red marbles are added to the container, after which the probability of randomly drawing a red marble is $\frac{2}{5}$. What was the total number of marbles originally in the container?

a. 14

b. **15** ✓

c. 21

d. 25

Let R, B, and G represent the initial number of red, blue, and green marbles, respectively. 1. From the given ratio $R:B=3:4$, we can express $R=3k$ and $B=4k$ for some positive integer k. The initial total number of marbles is $T=3k+4k+G=7k+G$. 2. When 6 green marbles are added: - The number of green marbles becomes $G+6$. - The new total number of marbles becomes $7k+G+6$. - The probability of drawing a green marble is $\frac{1}{3}$: $\frac{G+6}{7k+G+6}=\frac{1}{3}$ Cross-mul.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A list of 5 positive integers has a mean of 12, a median of 10, and a unique mode of 8. What is the greatest possible integer that could be in this list?

a. 17

b. 22

c. **23** ✓

d. 24

Let the 5 positive integers written in nondecreasing order be $a\le b\le c\le d\le e$. 1. **Median:** In an ordered set of 5 values, the median is the third value, so $c=10$. 2. **Unique Mode:** The mode is 8. Since $8<10$, 8 must occupy the positions before the median, so $a=8$ and $b=8$. Because 8 is the *unique* mode, no other value in the list can appear 2 or more times. 3. **Mean:** The mean of the 5 integers is 12, so their sum is: $a+b+c+d+e=5\times12=60$ Substituting $a=8$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A list consists of 7 distinct integers. The arithmetic mean of the 7 integers is 24, and the median is 22. If the 3 smallest integers are 15, 18, and 20, what is the maximum possible value of the largest integer in the list?

a. **46** ✓

b. 47

c. 48

d. 49

Let the 7 distinct integers in ascending order be $x_1 < x_2 < x_3 < x_4 < x_5 < x_6 < x_7$. 1. From the problem, the sum of all 7 integers is: $\text{Sum} = 7 \times 24 = 168$ 2. The median of a set of 7 ordered integers is the 4th integer, so $x_4 = 22$. 3. The 3 smallest integers are given as $x_1 = 15$, $x_2 = 18$, and $x_3 = 20$. 4. The sum of the first four integers is: $15 + 18 + 20 + 22 = 75$ 5. The sum of the remaining three integers must be: $x_5 + x_6 + x_7 = 168 - 75 = 93$ 6. To maximize x_7, we mus.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 20 balls numbered 1 to 20. One ball is drawn at random. What is the probability that the number drawn is at most 3?

a. $\frac{14}{25}$

b. $\frac{1}{2}$

c. 0

d. $\frac{3}{20}$ ✓

$P = \frac{\text{favorable}}{\text{total}} = \frac{3}{20} = \frac{3}{20}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

A bag has 7 red, 8 blue, and 2 green marbles. One marble is drawn at random. What is the probability it is blue?

a. $\frac{3}{8}$

b. $\frac{1}{5}$

c. $\frac{1}{3}$

d. $\frac{8}{17}$ ✓

$P = \frac{\text{favorable}}{\text{total}} = \frac{8}{17} = \frac{8}{17}$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

A bag has 3 red, 8 blue, and 3 green marbles. One marble is drawn at random. What is the probability it is red?

a. $\frac{13}{25}$

b. 1

c. $\frac{3}{16}$

d. $\frac{3}{14}$ ✓

$P = \frac{\text{favorable}}{\text{total}} = \frac{3}{14} = \frac{3}{14}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Plane Geometry

A standard six-sided die is rolled once. What is the probability of rolling 3 or 4?

a. $\frac{15}{16}$

b. $\frac{1}{4}$

c. $\frac{1}{3}$ ✓

d. 0

$P = \frac{\text{favorable}}{\text{total}} = \frac{2}{6} = \frac{1}{3}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

A standard six-sided die is rolled once. What is the probability of rolling a number less than 4?

a. $\frac{1}{5}$

b. $\frac{3}{4}$

c. $\frac{7}{8}$

d. $\frac{1}{2}$ ✓

$P = \frac{\text{favorable}}{\text{total}} = \frac{3}{6} = \frac{1}{2}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A standard six-sided die is rolled once. What is the probability of rolling 1 or 2 or 4 or 5?

a. $\frac{2}{3}$ ✓

b. $\frac{9}{10}$

c. $\frac{3}{5}$

d. $\frac{1}{4}$

$P = \frac{\text{favorable}}{\text{total}} = \frac{4}{6} = \frac{2}{3}$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

A class of 10 students averaged 62 on a test, and another class of 15 students averaged 77. What is the combined average?

a. 105

b. 158

c. **71** ✓

d. 122

Combined average $= \frac{10(62) + 15(77)}{10 + 15} = \frac{1775}{25} = 71$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the mode of the data set: 9, 17, 5, 6, 9, 20.

a. **9** ✓

b. 2

c. 33

d. 37

The value 9 appears most frequently (twice), so the mode is 9.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Functions & Graphs

If the probability of an event is $\frac{2}{15}$, what is the probability the event does NOT occur?

a. $\frac{13}{15}$ ✓

b. $\frac{9}{10}$

c. 0

d. $\frac{1}{8}$

$P(\text{not } A) = 1 - P(A) = 1 - \frac{2}{15} = \frac{13}{15}$.

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